

1 Highlights

2 **Surface Deformation Associated with the 2007 M_w 8.1 Solomon Is-** 3 **lands Earthquake: A Joint InSAR and Coral Geodesy Analysis**

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- 6 • A new method for separating deformation signals from tropospheric
7 noise is developed.
- 8 • InSAR captures deformation during and after the 2007 Solomon Islands
9 earthquake.
- 10 • Vertical deformation time series is reconstructed from field coral $\delta^{13}\text{C}$
11 measurements.
- 12 • Coral $\delta^{13}\text{C}$ measurements greatly extend surface deformation records
13 back in time.
- 14 • InSAR and coral observations during the postseismic period are con-
15 sistent.

16 Surface Deformation Associated with the 2007 M_w 8.1
17 Solomon Islands Earthquake: A Joint InSAR and
18 Coral Geodesy Analysis

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21 **Abstract**

Spaceborne Interferometric Synthetic Aperture Radar (InSAR) observations and $\delta^{13}\text{C}$ measurements from coral microatolls are utilized to investigate surface deformation associated with the 2007 M_w 8.1 Solomon Islands earthquake. Due to limited field validation data, a new InSAR analysis method is designed to distinguish deformation signals from tropospheric noise based on stacking solutions derived from subsets of interferograms. InSAR results reveal up to -235.8 cm of line-of-sight (LOS) deformation during the earthquake, as well as a LOS deformation rate decay from 26.4 to 5.9 cm/year within ~ 2 years after the earthquake. To further extend surface deformation records back in time, coral samples were collected from a *Porites lutea* microatoll grown in the western lagoon of Vonavona (locally known as Parara) Island. Vertical tectonic deformation time series (1996-2011) derived from coral Skeletal $\delta^{13}\text{C}$ measurements show a ~ 2.3 cm/year subsidence trend over the ten year period prior to the earthquake, a 76 cm uplift during the 2007 earthquake, and ~ 60 cm subsidence within the first 2.5 years after the earthquake. The characteristics of the temporal evolution of postseismic deformation inferred from independent coral and InSAR data are consistent. This is the first joint InSAR and coral analysis that spans the rupture and the pre- and postseismic phases of an earthquake cycle, and their complementary use expands the temporal and spatial coverage of surface deformation observations over this active subduction zone.

22 **Keywords:** InSAR, Coral Geodesy, Solomon Islands, Earthquakes

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1. Introduction

Understanding the earthquake cycle is crucial for investigating the potential for future hazards (Scholz, 2002; Savage and Thatcher, 1992). The pre-seismic period is characterized by strain accumulation due to the locking of the faults, which can last for tens of years. The coseismic period occurs as the accumulated stress exceeds the strength of the fault, and energy is released in the form of seismic waves in a few seconds or minutes. The postseismic period is defined by continued deformation stemming from both afterslip in the near field and various forms of relaxation in the far field over a few days to several months or years. Because the shallow megathrust (e.g., < 20 km depth) is typically located beneath the ocean, it is not always feasible to use land-based observations to study crustal deformation processes in such cases. However, the Solomon Islands, which was the site of a $M_w 8.1$ earthquake that occurred on April 1, 2007, is an exception to this. Here, islands are located within several kilometers of the trench, offering a rare opportunity to quantify crustal deformation processes on part of the shallow megathrust. Despite this unique geological setting, the Solomon Islands are a data-scarce environment characterized by large gaps in observational data, a lack of in situ measurements, and a remote and difficult-to-access site. There are no permanent GPS stations on the New Georgia island chain near the epicenter and only two in the entirety of the Solomon Islands. Limited campaign GPS data exist; but the duration of coverage is much shorter and the network is sparser than typical permanent networks.

Satellite remote sensing offers an opportunity to fill these observational gaps. Spaceborne Interferometric Synthetic Aperture Radar (InSAR) has been used to reconstruct millimeter to centimeter—surface deformations with tens to hundreds of meters of spatial resolution over broad areas (Massonnet et al., 1993). While InSAR has been successfully applied numerous times to investigate the earthquake cycle around the world (Wright et al., 2001; Walters et al., 2014; Delouis et al., 2002; Cakir et al., 2014; Kaneko et al., 2013; Lindsey et al., 2013; Lyons and Sandwell, 2003; Tong et al., 2013; Fialko et al., 2001; Johanson et al., 2005; Wei et al., 2015; Guns et al., 2022; Liu et al., 2018; Elliott et al., 2008; Bell et al., 2011; Wang and Wright, 2012; Wright et al., 2004; Ryder et al., 2007; Sijia et al., 2020; Walters et al., 2009; Lindsey et al., 2015), InSAR studies over the Solomon Islands are scarce.

58 Dense vegetation leads to severe phase decorrelation noise even in C-band
 59 Sentinel-1 interferograms with short temporal baselines (e.g., 12 days). While
 60 L-band InSAR data are less prone to decorrelation, ALOS PALSAR data are
 61 the only L-band InSAR data with open access. Because the number of ALOS
 62 PALSAR acquisitions are limited and irregular (~ 10 -15 SAR acquisitions
 63 over the entire mission duration of 2006-2010), it is very difficult to mitigate
 64 tropospheric turbulence noise (typically on the order of a few centimeters)
 65 due to the region's humid tropical climate. Furthermore, the availability
 66 of in situ measurements is extremely limited, making the validation of noisy
 67 InSAR results difficult. Miyagi et al. (2009) reported over 2 meters of LOS co-
 68 seismic deformation from the 2007 $M_w 8.1$ Solomon Islands earthquake based
 69 on ALOS PALSAR observations. Due to the difficulty in unwrapping in-
 70 terferograms over island regions, each of the three ALOS radar paths was
 71 processed individually. Miyagi et al. (2009) further solved for the slip distri-
 72 bution of this earthquake based on InSAR coseismic deformation estimates
 73 and limited field data, which suggested that the largest slip was around the
 74 hypocenter and centroid. Based on their slip model, they concluded that (1)
 75 the largest slip may be related to the subduction zone between the Wood-
 76 lark and Australian plates, and (2) the smaller magnitude slip between the
 77 hypocenter and the centroid may be attributed to thermal activity from the
 78 volcano located on Simbo Island. To our knowledge, this is the only pub-
 79 lished InSAR surface deformation study related to the 2007 Solomon Islands
 80 earthquake.

81 In this study, we show that L-band InSAR data can maintain phase coher-
 82 ence for ~ 2 years over the densely vegetated Solomon Islands. This makes it
 83 possible to detect both coseismic and postseismic deformation caused by the
 84 2007 Solomon Islands earthquake. Because postseismic deformation is much
 85 smaller than coseismic deformation, a major difficulty is the identification
 86 of InSAR phase signatures due to tropospheric noise that is comparable or
 87 larger than the signal of interest. Tropospheric noise consists of two compo-
 88 nents: (1) a stratified, topography-correlated component from variations in
 89 conditions in the vertical atmospheric column and (2) a turbulent and ran-
 90 dom component related to local weather fluctuations (Doin et al., 2009; Li
 91 et al., 2022). Although weather models and GNSS tropospheric zenith delay
 92 estimates have proven success in mitigating the stratified component (Doin
 93 et al., 2009; Yu et al., 2020), these external tropospheric correction layers of-
 94 ten lack the spatial and temporal resolution to capture the localized turbulent
 95 component. To overcome this challenge, a new InSAR analysis method is de-

96 veloped to identify InSAR deformation signals from tropospheric noise based
 97 on their distinct temporal and spatial characteristics revealed by subsets of
 98 interferogram stacks. The method works well even with a limited number of
 99 high-quality SAR scenes. To further fill the existing observation gaps, this
 100 study also aims to advance the methodology for paleo-geodesy through the
 101 analysis of $\delta^{13}\text{C}$ measurements of coral skeleton samples in a given Porites mi-
 102 croatoll located near the epicenter of the 2007 Solomon Islands earthquake.
 103 The vertical deformation time series inferred from both coral morphology
 104 and $\delta^{13}\text{C}$ geochemical analysis expands the temporal horizon of existing sur-
 105 face deformation studies back to 1996. This temporal expansion covers the
 106 interseismic period observational gap that exists with the ALOS data. This
 107 study is the first in which InSAR and coral measurements are utilized as
 108 complementary data sets.

109 **2. Background**

110 The Solomon Islands are situated astride an active plate boundary zone
 111 between the Australian and Pacific plates (Figure 1). Based on GPS cam-
 112 paigns, significant rates of oblique convergence ($\sim 10\text{-}11$ cm/year) occur be-
 113 tween these two plates (Demets et al., 1990; Wallace et al., 2004; Chen et al.,
 114 2011). Additionally, the Woodlark microplate subducts beneath the north
 115 Solomons and New Britain. This microplate rotates clockwise, pulling away
 116 from the Australian plate, and thus further modifies the kinematics and sub-
 117 duction rates at the New Britain and San Cristobal Trenches ($2.4^\circ/\text{Ma}$ for
 118 0.5 Ma to present) (Weissel et al., 1982; Benes et al., 1994; Wallace et al.,
 119 2004, 2014). The main boundary between the Pacific and Woodlark plates
 120 is the Woodlark basin, which is an active extensional basin where a new
 121 oceanic crust formation and seafloor spreading occurs, at rates of 4 to 5.6
 122 cm/year in the eastern Woodlark basin (Wallace et al., 2014). Subduction of
 123 the Australian and Woodlark Plates occurs at the San Cristobal trench just
 124 west of the Solomon Islands at rates of $\sim 9\text{-}10$ cm/year (Wallace et al., 2014)
 125 (Figure 1). The western Solomons, including the New Georgia island chain,
 126 are in close proximity to the San Cristobal trench. Ranongga, the island west
 127 of Kolombangara, is a mere 7 kilometers from the trench (Chen et al., 2009),
 128 and the western coast of Rendova is only ~ 20 kilometers from the trench.
 129 Interestingly, Simbo Island, just northwest of the island chain of interest, lies
 130 directly atop the subducting plate. These characteristics further solidify the
 131 Solomon Islands as an ideal laboratory for studying surface deformation from

132 spaceborne InSAR data.

133 Furthermore, the presence of coral microatolls fringing the islands pro-
134 vides an alternative way to reconstruct vertical deformation history in this
135 tropical region (Taylor et al., 1987). Coral species grow both upward and
136 horizontally; however, upward growth is limited by water depth (exposure
137 to air and direct sunlight). This limit is known as the highest level of sur-
138 vival (HLS). During sustained emergence above the ocean surface, exposed
139 coral surfaces die and produce notable die-downs in the coral morphology,
140 and growth continues laterally and upward at the HLS. When relative sea
141 level rises (or the ocean floor subsides), coral colonies grow vertically toward
142 the sea surface at rates ranging from 1 to 1.5 cm/yr. A striking example of
143 such growth and die-downs is provided by the $M_w 8.1$ earthquake of April 1,
144 2007, which produced dramatic vertical coastal displacements. Taylor et al.
145 (2008) utilized coral microatolls to investigate the catastrophic event that
146 ruptured just west of the New Georgia island chain. They analyzed the
147 vertical coral growth and die-downs to measure the associated coastal coseis-
148 mic subsidence and uplift, enabling some of the few coseismic deformation
149 measurements within several kilometers of the trench. Their field surveys
150 documented 0.73 ± 0.14 m of coseismic subsidence at Simbo Island, and up
151 to 2.46 ± 0.14 m of coseismic uplift at Ranongga Island, located only 8 km
152 away on the overriding plate. Their results revealed that the 2007 earthquake
153 ruptured across a triple junction where the Australian and Woodlark plates
154 subduct beneath the overriding Pacific plate. Similarly, coral microatolls
155 have been used to study coseismic vertical deformation in other regions that
156 have experienced large magnitude earthquakes, including Haiti and Sumatra
157 (Zachariasen et al., 1999; Briggs et al., 2006; Meltzner et al., 2010; Weil-
158 Accardo et al., 2016).

159 **3. Method**

160 *3.1. InSAR Data Processing*

161 ALOS PALSAR scenes acquired between December 28, 2006 and Jan-
162 uary 22, 2010 over the area of interest (Figure 2a) were downloaded from
163 the Alaska Satellite Facility (Table S1). All SAR scenes were processed us-
164 ing a motion-compensation imaging radar processor (Zebker et al., 2010),
165 and no spatial or temporal baseline constraints were applied during interfer-
166 ogram formation. All interferograms were multi-looked to 240-meter pixel

spacing, and the topographic phase component was removed using a Digital Elevation Model (DEM; 30 m pixel-spacing) derived from the Shuttle Radar Topography Mission (SRTM) (Farr et al., 2007). Interferograms acquired on the same day from the same path were merged by minimizing the phase difference between these overlapping regions of the adjacent frames, and the Goldstein filter was applied to merged interferograms to improve spatial phase coherence (Goldstein and Werner, 1998).

These interferograms were then unwrapped using Statistical-cost, Network-flow Algorithm for Phase Unwrapping (SNAPHU) algorithm (Chen and Zebker, 2001). SNAPHU attempts to unwrap InSAR phase measurements at all pixels including bodies of water where phase is completely decorrelated and random in space. This can dramatically increase computing time and introduce unwrapping errors at nearby land pixels. To overcome this challenge, land and ocean pixels were classified using an average phase coherence threshold of 0.11, and then wrapped phase values at ocean pixels were replaced by wrapped phase values at nearby land-pixels (e.g., Shanker and Zebker (2009)). As shown in Figure S1, the wrapped phase value ϕ_{ocean} at an ocean pixel of interest was replaced by the weighted sum of wrapped phase values at nearby land pixels as:

$$\phi_{ocean} = \frac{\sum w_i \phi_i}{\sum w_i} \quad (1)$$

where ϕ_i is the wrapped phase of the i^{th} land pixel within a square box of length 11 pixels centered at the ocean pixel, $w_i = 1/r_i$, and r_i is the distance between the ocean pixel of interest and the i^{th} land pixel within the square box. This phase filling process was performed for a second time to further extrapolate phase observations at land pixels into ocean area, which reduces large phase discontinuities at the coastline and phase unwrapping errors at land pixels. After unwrapping, the land-ocean mask was employed to identify land pixels for surface deformation analysis.

At a land pixel of interest, InSAR unwrapped phase ϕ measures surface deformation Δd_{LOS} along the radar line-of-sight direction with respect to a reference pixel as:

$$\phi \approx \frac{4\pi}{\lambda} \Delta d_{LOS} = \frac{4\pi}{\lambda} (e_1 \Delta d_e + e_2 \Delta d_n + e_3 \Delta d_u) \quad (2)$$

where λ is the radar wavelength. Δd_e , Δd_n , and Δd_u are surface deformation in east, north and vertical directions, and $\mathbf{e} = [e_1, e_2, e_3]$ is the

199 LOS unit vector. For the ALOS PALSAR data over the Solomon Islands,
 200 $\mathbf{e} = [0.6, 0.13, -0.79]$ at the mid-swath, and the look angle variation is
 201 not substantial across the ALOS PALSAR scenes. Coseismic deformation
 202 measurements (in east, north, and vertical directions) of the April 1, 2007
 203 Solomon Islands earthquake were recorded at three campaign GPS stations
 204 RAVA, MUND, and HUSU (Figure 2a). The GPS station RAVA (8.721°S
 205 157.407°E) was chosen as the reference point to calibrate the InSAR coseis-
 206 mic deformation map. The other two stations were used to validate InSAR
 207 results. Postseismic GPS surface deformation time series were recorded at
 208 MUND for a brief period from April 23, 2007 through May 5, 2007. How-
 209 ever, the limited data acquisition period makes it impossible to capture the
 210 postseismic deformation rate change over the ALOS PALSAR study period.
 211 Therefore, postseismic interferograms were not calibrated to campaign GPS
 212 data. Instead, InSAR postseismic deformation rate estimates were calcu-
 213 lated with respect to a local reference point (8.202°S 157.355°E) on the New
 214 Georgia island away from the epicenter and the coast.

215 *3.2. Coseismic and Postseismic LOS deformation solutions*

216 As shown in Equation (2), InSAR calculates the phase difference between
 217 two SAR acquisitions, from which the LOS surface deformation between these
 218 two acquisition dates can be derived (Rosen et al., 2000). Given that earth-
 219 quakes occur over a short period of seconds to minutes and the ALOS satellite
 220 revisit time is 46 days (the acquisitions were irregular with data gaps), a co-
 221 seismic deformation signal can be modeled as a step function (Figure 3a),
 222 and can be captured by interferograms that span the earthquake. Here one
 223 coseismic interferogram was formed using two Path 344 SAR scenes acquired
 224 on March 1, 2007 and April 16, 2007. Another coseismic interferogram was
 225 formed using two Path 343 SAR scenes acquired on February 12, 2007 and
 226 May 15, 2007. The two-frame interferograms were then merged to generate
 227 a single 4-frame coseismic interferogram. The magnitude of the expected
 228 coseismic deformation signal in the near-field of the event is on the order
 229 of meters (Miyagi et al., 2009; Taylor et al., 2008), which is well above the
 230 InSAR noise level. As a result, averaging multiple interferograms that span
 231 the April 1, 2007 Solomon Islands earthquake is not necessary.

232 The postseismic signal is expected to follow an exponential or logarithmic
 233 decay (Figure 3b) depending on the postseismic mechanism at play (Sobrero
 234 et al., 2020; Yusiyanti et al., 2023). Because the magnitude of the postseismic

235 deformation is significantly smaller than the coseismic deformation, a stack-
 236 ing approach (Sandwell and Price, 1998; Staniewicz et al., 2020) is employed
 237 to calculate the average LOS velocity, $\bar{v}_k$, for the k^{th} period of interest after
 238 the 2007 Solomon Islands earthquake:

$$S_k = \frac{\lambda}{4\pi} \frac{\sum_{i \in G_k} \phi_i}{\sum_{i \in G_k} \Delta t_i} = \frac{\sum_{i \in G_k} \Delta d_{LOS,i}}{\sum_{i \in G_k} \Delta t_i} + \delta = \bar{v}_k + \delta \quad (3)$$

239 where S_k denotes the k^{th} stack, $\lambda = 24$ cm is the ALOS PALSAR radar
 240 wavelength, and G_k is a set of interferograms that were formed using two
 241 SAR scenes acquired within the k^{th} period of interest. ϕ_i , $\Delta d_{LOS,i}$, and
 242 Δt_i are the unwrapped phase, the corresponding LOS deformation, and the
 243 temporal baseline of the i^{th} interferogram in G_k , respectively. δ is the residual
 244 noise term.

245 Nine ALOS PALSAR path 344 scenes acquired between April 16, 2007
 246 and January 19, 2010 were used in the postseismic deformation analysis (Fig-
 247 ure 3b). Here the postseismic deformation analysis is limited to every land
 248 pixel in path 344. This is because path 344 is closer to the 2007 earthquake
 249 epicenter than path 343, and the expected signal-to-noise ratio (SNR) is bet-
 250 ter. Following Equation (3), the average LOS velocity was calculated over
 251 three consecutive periods of interest: (1) April 16, 2007 through October
 252 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18,
 253 2008 through January 22, 2010. The interferograms used in each consecutive
 254 stack are listed in Table 1. Furthermore, the average LOS velocity was cal-
 255 culated over the following growing time windows starting on April 16, 2007
 256 through: (1) September 1, 2007 (2 acquisitions and 1 interferogram); (2)
 257 October 17, 2007 (3 acquisitions and 3 interferograms); (3) January 17, 2008
 258 (4 acquisitions and 6 interferograms); (4) March 3, 2008 (5 acquisitions and
 259 10 interferograms); (5) April 18, 2008 (6 acquisitions and 15 interferograms);
 260 (6) September 3, 2008 (7 acquisitions and 21 interferograms); (7) January
 261 19, 2009 (8 acquisitions and 28 interferograms); and (8) January 22, 2010 (9
 262 acquisitions and 36 interferograms). Section 3.3 further presents a method to
 263 determine whether postseismic deformation signals derived from these stacks
 264 of interferograms are above the residual noise level and thus detectable.

265 3.3. InSAR Measurement Uncertainty Quantification

266 The unwrapped InSAR phase at a pixel of interest, ϕ , can be written as:

$$\phi = \frac{4\pi}{\lambda} \Delta d_{LOS} + \Delta\phi_{DEM} + \Delta\phi_{decorr} + \Delta\phi_{unwrap} + \Delta\phi_{orb} + \Delta\phi_{iono} + \Delta\phi_{tropo} + \Delta\phi_n \quad (4)$$

267 where λ is the radar wavelength, and Δd_{LOS} is the LOS deformation between
 268 the two SAR acquisitions used to form the interferogram. The remaining
 269 terms are the result of various error sources: inaccurate DEM ($\Delta\phi_{dem}$), phase
 270 decorrelation ($\Delta\phi_{decorr}$) and the resulting unwrapping errors ($\Delta\phi_{unwrap}$), or-
 271 bital errors ($\Delta\phi_{orb}$), ionospheric ($\Delta\phi_{iono}$) and tropospheric ($\Delta\phi_{tropo}$) artifacts,
 272 and other smaller error terms ($\Delta\phi_n$) such as changes in soil moisture.

273 Given that the magnitude of the observed InSAR phase does not correlate
 274 with InSAR perpendicular baseline, a key signature of DEM errors (Fattahi
 275 and Amelung, 2013), the impact of DEM errors is minimal. Based on vi-
 276 sual inspection, all interferograms formed using path 344 scenes acquired
 277 on two dates (January 14, 2007 and October 19, 2008) suffer from severe
 278 phase decorrelation artifacts and relatively large unwrapping errors. This is
 279 likely due to changes in surface scattering properties on these two days (e.g.,
 280 due to a severe weather event). These two SAR acquisitions and the associ-
 281 ated interferograms were excluded, and a total of nine high-quality path 344
 282 SAR acquisitions were used in the postseismic deformation analysis (Figure
 283 3b). For all interferograms formed using these nine path 344 acquisitions,
 284 InSAR phase coherence is well maintained regardless of temporal baselines,
 285 except over Tetepare Island, which lies southeast of Rendova. Consequently,
 286 the postseismic deformation analysis focuses on Vonavona (locally known as
 287 Parara), Kohinggo, Kolombangara, and New Georgia, where decorrelation
 288 artifacts $\Delta\phi_{decorr}$ and unwrapping errors ($\Delta\phi_{unwrap}$) are not substantial. A
 289 planar phase ramp was estimated and removed in each postseismic interfero-
 290 gram to account for orbital errors ($\Delta\phi_{orb}$) and long-wavelength tropospheric
 291 and ionospheric noise. Because the orientation of the observed phase ramp in
 292 each interferogram varies, these ramps are unlikely related to true deforma-
 293 tion signals. The other error sources, $\Delta\phi_n$, associated with thermal and soil
 294 moisture effects, are on the order of millimeters and much smaller than other
 295 noise terms (Michaelides et al., 2019). Thus, it follows that the dominant
 296 error term is tropospheric noise in the Solomon Islands ALOS PALSAR data
 297 set, and Equation (4) can be simplified to:

$$\phi_{ij} = \frac{4\pi}{\lambda} (\Delta d_{ij} + n_j - n_i) \quad (5)$$

where λ is the radar wavelength and Δd_{ij} is the LOS surface deformation between two SAR acquisitions acquired on dates i and j . The tropospheric noise in the LOS direction on each acquisition date is represented as n_i and n_j , respectively.

Based on nine ALOS PALSAR path 344 scenes (Figure 3b), consider the stacking results over three consecutive periods of interest: (1) April 16, 2007 through October 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18, 2008 through January 22, 2010. Following Equation (3) and Equation (5), each of the three stacks defined in Table 1 can be expressed as:

$$S_1 = \bar{v}_1 + \frac{2n_3 - 2n_1}{\sum_{i=1, i < j}^3 \Delta t_{ij}} \quad (6)$$

$$S_2 = \bar{v}_2 + \frac{3n_6 + n_5 - n_4 - 3n_3}{\sum_{i=3, i < j}^6 \Delta t_{ij}} \quad (7)$$

$$S_3 = \bar{v}_3 + \frac{3n_9 + n_8 - n_7 - 3n_6}{\sum_{i=6, i < j}^9 \Delta t_{ij}} \quad (8)$$

Here $\bar{v}_1$, $\bar{v}_2$, and $\bar{v}_3$ are the average deformation rates for each of the three periods of interest. n_i is the LOS tropospheric noise of the i^{th} SAR acquisition ($i = 1, \dots, 9$), and the total time span of all interferograms ($\sum \Delta t_{ij}$) in the three stacks are 1.01, 1.64, and 5.67 years, respectively. In the case where postseismic deformation signals are well above the noise level, a consistent deformation pattern should be observed in all three stacking solutions. At a given location, the observed deformation rate is expected to decrease in time (e.g. a larger rate is expected in S_1 than S_3). For a given stack solution S_k , the observed deformation rate is expected to decrease as the distance to the epicenter increases. In contrast, tropospheric errors are not correlated for time increments greater than one day (Zebker et al., 1997; Emardson et al., 2003). Given that the ALOS PALSAR satellite revisit cycle is 46 days, $n_1, \dots, n_9$ are temporally uncorrelated. This suggests that the residual tropospheric noise patterns in S_1 and S_3 are not correlated. The residual noise level in S_3 tends to be smaller than in S_1 because the total time span of all interferograms in S_3 (5.67 years) is more than five times larger than in S_1 (1.01 years). Furthermore, the third SAR acquisition (October 17, 2017)

was used to form interferograms in both S_1 and S_2 . As shown in Equation (6) and Equation (8), the contribution of n_3 in S_1 and S_2 has the opposite sign. Similarly, the contribution of n_6 in S_2 and S_3 also has the opposite sign. In summary, the temporal and spatial characteristics of tropospheric noise and postseismic deformation signals are different. This allows determination of whether postseismic deformation signals are detectable by a comparison of S_1 , S_2 , and S_3 .

Next, consider the following growing time windows starting on April 16, 2007 through: (1) September 1, 2007 (2 acquisitions); (2) October 17, 2007 (3 acquisitions); (3) January 17, 2008 (4 acquisitions); (4) March 3, 2008 (5 acquisitions); (5) April 18, 2008 (6 acquisitions); (6) September 3, 2008 (7 acquisitions); (7) January 19, 2009 (8 acquisitions); and (8) January 22, 2010 (9 acquisitions). For the case of M SAR acquisitions that all $N = M \times (M - 1)/2$ interferograms are formed, the stacking result, S_M , can be written as:

$$S_M = \bar{v}_M + \frac{1}{\sum_i \Delta t_i} \left(\sum_m^M w_m n_m \right) \quad (9)$$

where $\bar{v}_M$ is the average deformation rates for the M^{th} time window, and $\sum_i \Delta t_i$ is the total time span of all interferograms in the subset. n_m is the tropospheric noise of the m^{th} SAR acquisition, and the weight applied to the tropospheric noise of the m^{th} SAR scene, w_m , can be described as

$$w_m = 2m - (M + 1) \quad (10)$$

Equation (9) and Equation (10) quantitatively describe the stacking results of eight growing time windows, where each stack contains an additional SAR scene forward in time. For $M = 2$, $w_1 = -1$ and $w_2 = 1$. For $M = 9$, the largest absolute weights are $w_1 = -8$ and $w_9 = 8$. Furthermore, as M increases, the total time span of all interferograms increases. For the Solomon Islands case, the total time space for $M = 2$ is 0.378 years, while the total time space for $M = 9$ is 34.5 years. This means that the residual tropospheric noise from the 1st SAR acquisition (April 16, 2007) is more than 11 times larger when $M = 2$ than when $M = 9$. In this growing time window stacking design, both the postseismic deformation signal and n_1 are expected to produce a spatially consistent pattern with reduced magnitude over time. In contrast, residual tropospheric noise from the remaining SAR acquisitions

is expected to produce random patterns in these eight stacking results. To further separate the contribution from the postseismic deformation signal and n_1 , an alternative stacking test excluding all interferograms formed using the 1st SAR acquisition (April 16, 2007) can be calculated, where n_1 does not contribute to the result.

3.4. Time Series of Vertical Displacements Derived from Coral $\delta^{13}\text{C}$ Measurements

The ALOS PALSAR mission only captured two acquisitions prior to the 2007 earthquake, and one of these two was excluded in InSAR analysis due to severe phase decorrelation. Consequently, years of the interseismic period cannot be reconstructed from InSAR data alone. In order to fill the interseismic data gap, this study further investigates the $\delta^{13}\text{C}$ ($^{13}\text{C}/^{12}\text{C}$) signal preserved in shallow-living coral microatolls and its ability to expand the temporal horizon of surface deformation records to nearly two decades. Vertical deformation derived from $\delta^{13}\text{C}$ measurements tends to be more accurate than coral morphology observations. This is because microatolls do not always emerge above the water during sea-level drops, and growth may not keep pace with rapid sea-level rise (Philibosian, 2025).

Although $\delta^{13}\text{C}$ variations in coral skeletons depend on several parameters, previous studies have established a linear relationship between coral skeletal $\delta^{13}\text{C}$ and water depth, with decreasing $\delta^{13}\text{C}$ values at greater depth. This is attributed to reduced light penetration, lower photosynthetic activity, and declining irradiance with depth (Weber and Woodhead, 1970; Weber et al., 1976; Swart, 1983). Based on this relationship, $\delta^{13}\text{C}$ acts as a proxy for relative sea level (RSL), which is influenced by both tectonic activity and climate patterns. Here climate signals often contain annual variations and some semblance of cycles (e.g., El Nino–Southern Oscillation), whereas tectonic signals behave differently – interseismic deformation often appears as a long-term trend in time series, and postseismic deformation often follows the expected exponential decay model. Furthermore, the relative sea level history can be compared with independent measurements, such as tide gauge data, to further isolate vertical deformation due to tectonic activity (Gagan et al., 2008, 2009).

The coral cross-section used in this study (Figure S2) was retrieved in April 2012 from a *Porites lutea* microatoll that had grown in the western lagoon of Vonavona (locally known as Parara) Island in the Solomon Islands

391 (8.221° S, 157.005° E). The coral slab measured $138 \times 100 \times 6$ cm, and its mor-
 392 phology records multiple exposed surfaces associated with emergence events
 393 during relative sea-level drops. All exposed surface measurements for this
 394 study include both the length of the dead surface and the depth of the wa-
 395 ter above the coral top at the time of collection. X-ray scans of the coral
 396 slab (Figure 4) reveal clear annual growth rings, analogous to tree rings.
 397 By counting the growth rings backward into time starting from the date of
 398 collection, historic events can be dated with annual or better temporal res-
 399 olution (Buddemeier and Taylor, 2000; Biemiller et al., 2020). For example,
 400 the April 2007 $M_w 8.1$ megathrust earthquake caused an uplift that killed the
 401 living coral surface (marked by the green line in Figure 4) (Taylor et al.,
 402 2008). Following this 2007 exposed surface line, samples were collected using
 403 a computer-controlled drill along 7 transects (A–G at different water depths
 404 prior to the 2007 earthquake) (marked by the yellow lines in Figure 4), each
 405 perpendicular to the growth rings. Each transect consisted of a series of
 406 about 10 single samples spanning a total of 50–60 mm in vertical direction
 407 (~ 3 years of coral growth). A total of 68 samples from Transects A–G, each
 408 with a drilled volume of $1.5 \times 1.5 \times 1.0$ mm, were obtained, and $\delta^{13}\text{C}$ were
 409 analyzed and recorded to estimate the linear relationship between $\delta^{13}\text{C}$ and
 410 water depth.

411 To further reconstruct $\delta^{13}\text{C}$ time series over longer time scales, a hori-
 412 zontal transect (H) spanning the years 1996–2011 (marked by the red line in
 413 Figure 4) was sampled and analyzed. A total of 118 samples along transect
 414 H enables reconstruction of inter-, co-, and postseismic vertical motion of
 415 the microatoll using the linear $\delta^{13}\text{C}$ -water depth relationship derived from
 416 the vertical sampling analysis. To isolate skeletal $\delta^{13}\text{C}$ signals associated
 417 with climate and tectonics, GPS-corrected tide-gauge records from Honiara
 418 (9.429° S, 159.955° E; Fig. 2b) were utilized as a control (Caldwell et al.,
 419 2001). Sea surface height measured by the satellites using radar altimetry
 420 (Zlotnicki et al., 2019) reveals a strong correlation between the study site
 421 with the tide gauge in Honiara and Bougainville. This means that the ab-
 422 solute sea levels at Vonavona and Honiara are similar. However, Honiara
 423 is too far away from the epicenter, and deformation signals related to the
 424 2007 earthquake cycle are not detectable here. The relative sea level history
 425 derived from the $\delta^{13}\text{C}$ measurements acts as a complementary data set that
 426 shows consistency with remote sensing observations. It is important to note
 427 that InSAR LOS measurements cannot be directly compared to vertical mea-
 428 surements from coral samples acquired at a different location. Nonetheless,

429 consistency in the temporal evolution of the deformation and the order of
430 magnitude can be analyzed as an alternative way of validation.

431 4. Results and Discussion

432 4.1. Coseismic Deformation Determined from InSAR

433 The 2007 Solomon Islands earthquake deformation map derived from the
434 ALOS PALSAR coseismic interferograms is shown on a circular color scale
435 in Figure 5. Here each fringe cycle (e.g., green to pink to yellow to green)
436 indicates 20 cm of positive LOS deformation. In this study, the LOS di-
437 rection is defined as from the satellite to the ground. Therefore, uplift or
438 westward motion leads to negative LOS deformation in ALOS PALSAR as-
439 cending geometry, whereas subsidence or eastward motion leads to positive
440 LOS deformation in the same geometry. Spatially continuous fringe patterns
441 are observed across multiple islands. This suggests that the proposed phase-
442 filling strategy can unwrap InSAR phase measurements over land surface
443 pixels isolated by ocean pixels successfully. The observed fringes over Von-
444 avona and Kohingo are the densest in space, because these two islands are
445 closest to the epicenter (Figure 2). The same coseismic LOS deformation map
446 is also presented on a continuous color map (Figure 5b), which highlights a
447 maximum magnitude of -235.8 centimeters of LOS deformation measured on
448 the southwestern coast of the Vonavona island, approximately 26 kilometers
449 from the epicenter. Kohingo, another western island near the epicenter,
450 experienced LOS deformation ranging from -100.1 to -194.4 cm. In contrast,
451 Kolombangara recorded LOS deformation ranging from -48.2 to -128.6 cm,
452 and Rendova recorded LOS deformation ranging from 11.8 to -69.5 cm. The
453 spatial deformation gradient is visible on the large New Georgia island. The
454 northwestern portion of the island near Kohingo experienced up to -123.4
455 centimeters of negative LOS deformation during the earthquake, while the
456 southeastern region experienced up to 18.6 centimeters positive LOS defor-
457 mation. Vangunu, the island farthest east in the study region, measured up
458 to 25.6 centimeters positive LOS deformation. This value is of significantly
459 smaller magnitude than that of Vonavona, as expected because of the loss of
460 energy in the seismic waves as they travel farther away from the epicenter.

461 Three GPS campaign stations are located within the coseismic map in
462 Figure 5: MUND, RAVA, and HUSU. RAVA was used as the reference point
463 to calibrate InSAR coseismic map, and the misfits between the InSAR LOS

464 and the MUND and HUSU GPS stations are 3.2 and -3.9 centimeters, re-
 465 spectively (Table 2). The misfit values are within the expected tropospheric
 466 noise range of 2 to 5 centimeters of a single interferogram over humid tropical
 467 islands (Chen et al., 2014). Additionally, the coseismic interferograms gen-
 468 erated from different paths acquired on different dates were merged without
 469 any visual discontinuities across path boundaries. This further confirms that
 470 the coseismic deformation signal is well above the noise level. Previous re-
 471 search reported up to 2.46 ± 0.14 m of coastal coseismic uplift at Ranongga
 472 island based on surveys of coral reef die-downs (Taylor et al., 2008). In com-
 473 parison, the maximum magnitude LOS deformation at the nearby Vonavona
 474 island is 235.8 centimeters. The order of the deformation (meter-level) and
 475 the spatial patterns are consistent between spaceborne InSAR and field coral
 476 observations. We note that it is not possible to validate InSAR co-seismic
 477 deformation estimates directly using field observations. This is because (1)
 478 InSAR measures surface deformation along the LOS direction, while coral
 479 surveys measure vertical deformation. Here the decomposition of InSAR
 480 LOS deformation into horizontal and vertical components is not possible due
 481 to the lack of multiple LOS imaging geometries; and (2) coral grows in the
 482 ocean, and InSAR can only measure surface deformation at land pixels. Thus
 483 there are no co-located InSAR and coral measurements.

484 *4.2. Postseismic Deformation Determined from InSAR*

485 Figure 6 shows the average LOS rate (in cm/year) over three consecutive
 486 time periods of interest: (a) the first six months following the earthquake
 487 from April through October 2007, (b) the following seven-month period cov-
 488 ering October 2007 through April 2008, and (c) the remaining period for
 489 which SAR acquisitions are available that spans April 2008 through January
 490 2010. In the green dashed box, the maximum LOS rate is observed near the
 491 epicenter on Vonavona with a similar spatial pattern for all three time peri-
 492 ods. At a given location, the observed LOS rate decreases over time. This is
 493 consistent with the temporal characteristics of the postseismic deformation
 494 signals. In contrast, the LOS patterns over the island regions outside the
 495 green dashed box are correlated in space but random in time – characteris-
 496 tics that align with tropospheric turbulence noise behavior as discussed in
 497 Section 3.3.

498 Figure 7 shows the average LOS deformation rates of eight overlapping
 499 time windows starting on April 16, 2007 through: (1) September 1, 2007;
 500 (2) October 17, 2007; (3) January 17, 2008; (4) March 3, 2008; (5) April

18, 2008; (6) September 3, 2008; (7) January 19, 2009; and (8) January 22, 2010. The first stack (Figure 7a) includes only a single interferogram (2007/04/16-2007/09/01), which contains the largest postseismic deformation signal during the first 5 months after the earthquake. Inside the green dashed box (over Vonavona), the observed LOS rate decreases over time (Figures 7(b)-(h)), and the spatial patterns (larger deformation rates are at pixels closer to the epicenter) are consistent in all eight stacks. Based on Equation (9) and Equation (10), both the postseismic deformation signal and tropospheric noise on the first SAR acquisition (April 16, 2007) may produce the pattern inside the green dashed box in Figure 7. Nonetheless, Figure 6(b) - (c) also contain this spatially consistent LOS pattern inside the green dashed box, and these two stacks do not include interferograms formed using the first SAR acquisition. Therefore, the observed LOS pattern over Vonavona is likely associated with postseismic deformation signals rather than tropospheric noise. On the other hand, the observed LOS pattern over the other islands –Kolombangara, Kohinggo, and New Georgia—are correlated in space but random in time. This is consistent with the finding of Figure 6 that there are no detectable signals above tropospheric noise level here. As shown in Section 3.3, the residual tropospheric noise decreases as the total time span of an interferogram stack increases (Figures 7(a)-(h)).

Based on ALOS InSAR stacking solutions, the average LOS rate on Vonavona from April 16, 2007 through (1) September 1, 2007, (2) October 17, 2007, (3) January 17, 2008, (4) March 3, 2008, (5) April 18, 2008, (6) September 3, 2008 are 26.4 cm/year, 19.5 cm/year, 14.2 cm/year, 12.7 cm/year, 10.5 cm/year, and 6.2 cm/year, respectively. The average LOS rate over two longer periods, spanning April 16, 2007 through January 19, 2009 and January 22, 2010, are both 5.9 cm/year. Our results follow the expected exponential decay behavior associated with postseismic afterslip (Perfettini and Avouac, 2004), which highlights the effectiveness of our algorithm for capturing the temporal evolution of surface deformation from a sparse InSAR data set.

4.3. Vertical deformation from Coral $\delta^{13}C$ Time Series

The paleo geodesy coral analysis focuses on the 1996–2011 period, which includes three phases: (1) the preseismic period prior to the 2007 earthquake, (2) coseismic emergence during the earthquake, and (3) the postseismic period after April 1, 2007. The morphology between years 1997 and 2007 suggests that the microatoll subsided a minimum of ~ 17 cm in a 10-year

538 period, was uplifted at least ~ 63 cm during the earthquake, and subse-
 539 quently subsided again by tens of centimeters (Figure 4). The coral scan in
 540 Figure 4 indicates a coseismic uplift extent of ~ 73 cm, which is derived by
 541 combining the vertical extent of the 2007 die-down surface (63 cm; marked
 542 by the green line in Figure 4) with the measured water depth (10 cm) above
 543 the coral top at the time of collection. The corresponding change in $\delta^{13}\text{C}$ is
 544 2‰, yielding an estimated $\delta^{13}\text{C}$ -water depth relationship of ~ 36.5 cm/‰
 545 from the morphology. Furthermore, coral samples from Transects A-G were
 546 used to quantify the relationship between water depth and $\delta^{13}\text{C}$. The results
 547 (Figure 8) confirm a linear trend as reported by previous studies (Weber
 548 and Woodhead, 1970; Weber et al., 1976; Swart, 1983): skeletal $\delta^{13}\text{C}$ values
 549 decrease with increasing water depth. The root-mean-square error (RMSE)
 550 indicates additional variability, which is likely associated with sub-annual
 551 to annual sea-level changes. Nonetheless, the mean slope of 37.67 cm/‰
 552 closely matches the coseismic estimate of 36.5 cm/‰ derived from the coral
 553 morphology, confirming that $\delta^{13}\text{C}$ is a reliable proxy for relative sea-level
 554 reconstruction. **Section S1 (Supporting Information) provides further dis-**
 555 **cussion of the difference between the mean estimates derived from $\delta^{13}\text{C}$ and**
 556 **morphology, and uncertainty associated with the coral measurements.**

557 The three phases of the earthquake cycle were further investigated through
 558 the relative sea level (RSL) history reconstructed from the $\delta^{13}\text{C}$ measure-
 559 ments from Transect H and the water-depth relationship derived from ver-
 560 tical sampling (Figure 8). The unfiltered mean relative sea level time series
 561 derived from the $\delta^{13}\text{C}$ measurements of coral samples is shown in Figure 9a.
 562 A clear vertical displacement, indicated by the sharp change in relative sea
 563 level, is present during the time of the 2007 Solomon Islands earthquake.
 564 This coseismic uplift indicated from the $\delta^{13}\text{C}$ is approximately 76 ± 10 cm.
 565 A vertical measurement of the visible die-down in the morphology a nearby
 566 coral sampling site (8.227°S, 157.012°E) from a previous study (Taylor et al.,
 567 2008) recorded uplift of 50 ± 15 cm, highlighting the agreement between the
 568 two in situ measurement methods. An L1 linear regression is applied to the
 569 time series prior to the 2007 earthquake, which reveal an approximately 2.3
 570 centimeters per year subsidence trend over the time period 1996 and 2007.
 571 Both the 2007 uplift and over a decade of subsidence are likely due to tec-
 572 tonic deformation rather than absolute sea level changes. This is because
 573 both signals are not present in the RSL time series recorded in the Honiara
 574 tide gauge data Figure 9b. The two RSL data sets both capture similar sea-
 575 sonal weather signals and climate events (e.g. El Nino). However, Honiara

576 is far away from the epicenter and the tectonic signals related to the 2007
577 earthquake cycle are not detectable here.

578 Following the coseismic uplift, coral time series demonstrates exponential
579 decay behavior during the post-seismic period, which is not present in the
580 Honiara tide gauge data. In Figure 10, this post-seismic signal is plotted
581 with the InSAR LOS deformation over the three consecutive time windows
582 discussed above in Section 4.2. The LOS deformation rate over each time
583 window of interest is known from Equation (9). The cumulative deforma-
584 tion plotted at the final date of each window was computed by multiplying
585 the LOS deformation rate of the window by the window’s time duration and
586 subsequently summing the value with the deformation from previous win-
587 dows. The similar temporal evolution between the LOS deformation and
588 the coral-derived relative sea level highlights the consistency between the
589 two independent data sets. This agreement among the $\delta^{13}\text{C}$ measurements
590 and remote sensing observations supports the conclusions drawn from the
591 InSAR-derived deformation.

592 5. Conclusion

593 In this study, surface deformation prior to, during, and after the 2007
594 Solomon Islands earthquake was reconstructed using InSAR and coral geodesy.
595 Phase filling allowed multiple islands from two separate ALOS PALSAR
596 paths to be successfully unwrapped as a single interferogram without dis-
597 continuities or visible unwrapping errors. The coseismic InSAR map was
598 calibrated and validated using coseismic displacements from three campaign
599 GPS stations, and is consistent with previously published results (Miyagi
600 et al., 2009). A new InSAR analysis method based on stacking subset of in-
601 terferograms was developed. This enabled, for the first time, the generation
602 of a postseismic InSAR time series of the 2007 Solomon Islands earthquake.
603 $\delta^{13}\text{C}$ measurements from a coral sample near Vonavona provided a record
604 of the relative sea level in the region, which contained a signal from both
605 oceanographic changes and vertical tectonic motion. Using a control data
606 set from a nearby tide gauge, the contribution from the absolute sea level
607 changes was quantified. Results suggested that the coral time series captured
608 an interseismic subsidence trend, a 76 cm coseismic uplift, and a decay of
609 subsidence rates following the earthquake. The characteristics of the tempo-
610 ral evolution of postseismic deformation inferred from independent coral and
611 InSAR data are consistent. While our InSAR analysis method works with a

612 limited number of SAR data sets—only 10 high-quality ALOS PALSAR scenes
613 are available—this method can be designed for robust uncertainty quantifica-
614 tion when a large number of SAR scenes become available (Zebker et al.,
615 2023; Zebker and Chen, 2024). The study highlights the potential of L-band
616 InSAR data for observing surface deformation over densely vegetated and
617 humid regions. With the combination of an L-band sensor for penetrating
618 dense vegetation and the shorter revisit time of 12 days, NISAR will produce
619 high-quality InSAR observations over under-surveyed subduction zones. Fur-
620 thermore, the results from the coral $\delta^{13}\text{C}$ measurements show promise as a
621 method to greatly expand the time horizon of vertical deformation history
622 and complement modern geodesy observations.

623 **Authorship Contribution Statement**

624 Conceptualization: The project was designed by J. Chen, L. Lavier, L.
625 Wallace, and F. Taylor. InSAR Analysis: L. Kendall and J. Chen. Coral
626 Analysis: L. Kendall, L. Lavier, F. Taylor and M.E. Karaesmen. Field data
627 collection: F. Taylor, L. Lavier, and L. Wallace. Writing—original draft prepa-
628 ration: L. Kendall. Writing—review and editing: all authors. Visualization:
629 L. Kendall and L. Wallace. Funding acquisition: L. Kendall, J. Chen, L.
630 Lavier, F. Taylor and L. Wallace.

631 **Open Research Section**

632 ALOS PALSAR SAR imagery over the New Georgia island chain in the
633 Solomon Islands (Paths 343 and 344, Frames 7010 and 7020) can be down-
634 loaded from the Alaska Satellite Facility at <https://search.asf.alaska.edu>. All
635 interferograms in this study were processed using a motion-compensation
636 imaging radar processor (Zebker et al., 2010). Comparable interferograms
637 can be generated with the following open-source InSAR packages: (1) In-
638 SAR Scientific Computing Environment 3 (ISCE3) (Rosen et al., 2018) or
639 (2) GMTSAR (Sandwell et al., 2011).

640 **Conflict of Interest**

641 The authors declare there are no conflicts of interest for this manuscript.

642 **6. Acknowledgments**

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653 and samples from their coasts.

Table 1: Interferograms used for estimating the average postseismic deformation rate over three consecutive periods of interest

S_1	S_2	S_3
April 2007 - October 2007	October 2007 - April 2008	April 2008 - January 2010
04/16/2007-09/01/2007	10/17/2007-01/17/2008	04/18/2008-09/03/2008
04/16/2007-10/17/2007	10/17/2007-03/03/2008	04/18/2008-01/19/2009
09/01/2007-10/17/2007	10/17/2007-04/18/2008	04/18/2008-01/22/2010
	01/17/2008-03/03/2008	09/03/2008-01/19/2009
	01/17/2008-04/18/2008	09/03/2008-01/22/2010
	03/03/2008-04/18/2008	01/19/2009-01/22/2010

Table 2: GPS Validation for InSAR-Derived coseismic Deformation Map

Station	Location	InSAR LOS (cm)	GPS LOS (cm)	Misfit (cm)	Distance from RA
RAVA	8.72°S, 157.41°E	7.17	7.17	0	0
MUND	8.33°S, 157.27°E	-86	-89.17	3.17	46.1
HUSU	8.50°S, 157.36°E	-37.54	-33.68	-3.86	25.1

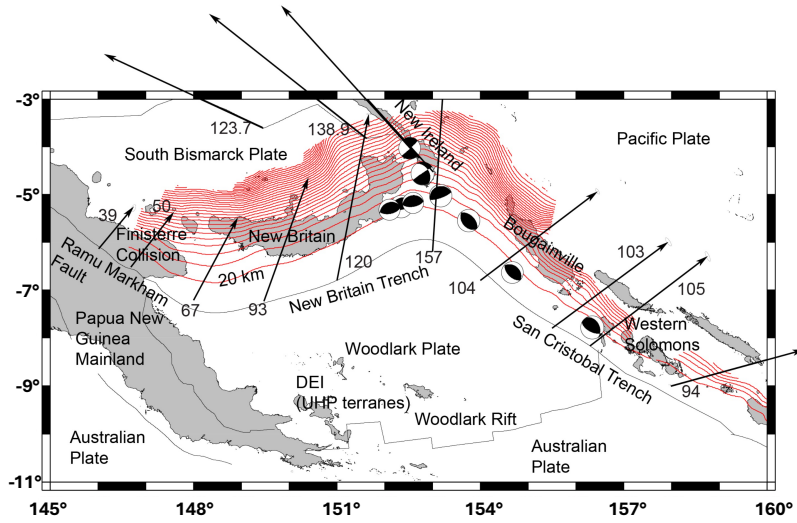


Figure 1: Convergence rates (black arrows, in mm/yr) and slab configuration (red contours) at the Ramu-Markham Fault, New Britain Trench, and San Cristobal Trench. Strike slip rates on the northern boundary of the South Bismarck Plate (e.g. on the Weitin fault that transects New Ireland) are also shown. Beach balls show focal mechanisms for earthquakes larger than Mw 7.5, 1999-present (from Global CMT catalog). New Britain slab contours are in 20 km intervals, with the maximum depth shown at 600 km (contours from Hayes et al. (2012)). West of 153°E, New Britain Trench rates are determined using Woodlark/South Bismarck Plate relative motion (Wallace et al., 2004), while east of 153°E rates are from Woodlark/Pacific Plate relative motion (Wallace et al., 2014). DEI stands for D’Entrecasteaux Island, and UHP stands for Ultra-High Pressure.

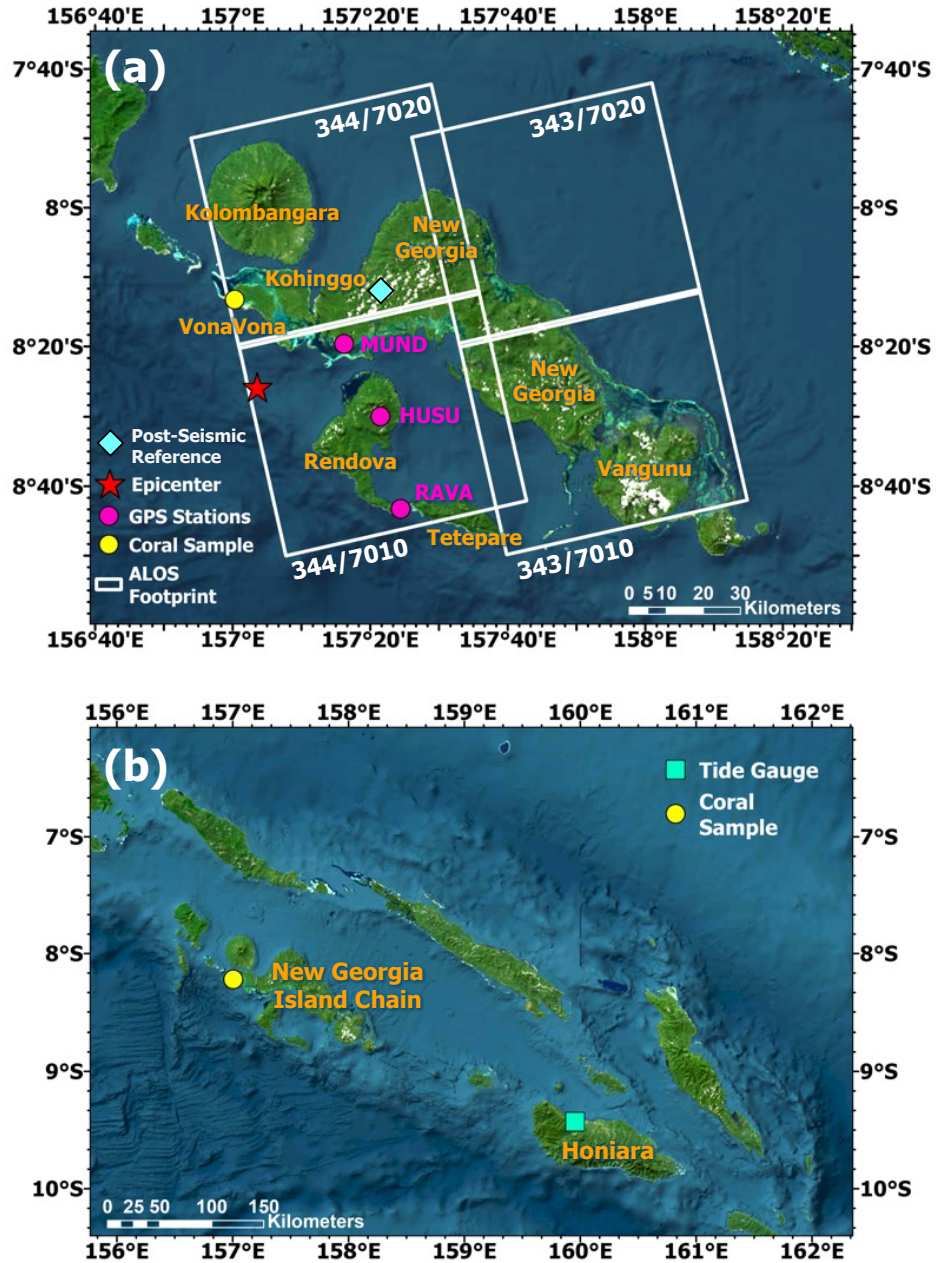


Figure 2: (a) Map of the study area over the Solomon Islands, specifically the New Georgia Island Chain. ALOS PALSAR footprints are outlined in white, and the path/frame numbers are labeled. Locations of three GPS stations, the coral sample site, and the epicenter of the April 1, 2007 Solomon Islands earthquake are labeled according to the legend. (b) The location of the coral sample site near VonaVona in the New Georgia Island chain and the location of the tide gauge off the coast of Honiara.

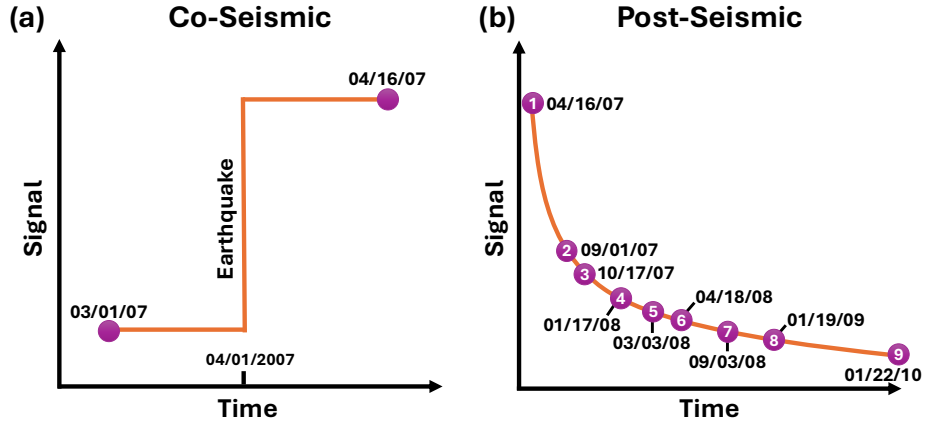


Figure 3: (a) The coseismic signal is modeled as a step function. This transient deformation can be captured by two SAR acquisitions that span the earthquake. (b) The postseismic deformation is expected to follow an exponential or logarithmic decay curve. High-quality Path 344 ALOS PALSAR acquisitions used in the postseismic deformation analysis are marked as purple dots. Note that data gaps are present and the time interval between two nearest acquisitions varies.

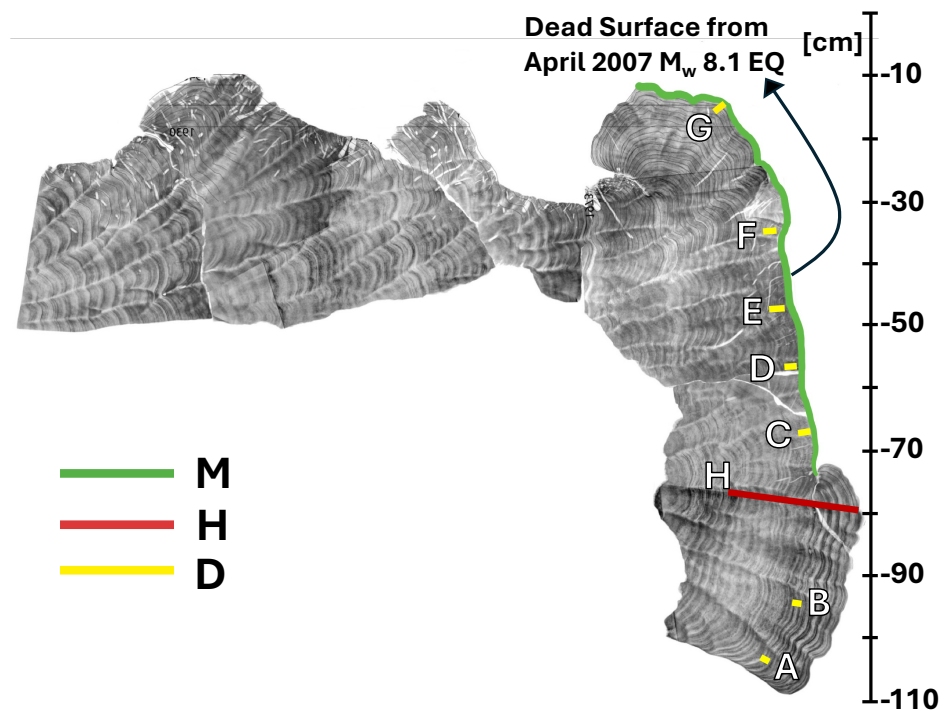


Figure 4: X-ray radiograph of a ~ 1 cm-thick slice of fossil *Porites* coral collected in 2012, five years after the April 2007 M_w8.1 earthquake. Growth bands reveal vertical extension rates of 1–1.5 cm/yr. Vertical transects (A–G) were drilled at 1.5 mm spacing to establish the $\delta^{13}\text{C}$ -depth relationship (marked by the yellow lines with the legend 'D' for depth). Transect H, whose samples were utilized for relative sea level reconstruction between 1996 and 2011, is highlighted by the red line. The notable die-down of the coral surface caused by a large uplift from the 2007 earthquake is highlighted by the green outline, labeled 'M' for morphology.

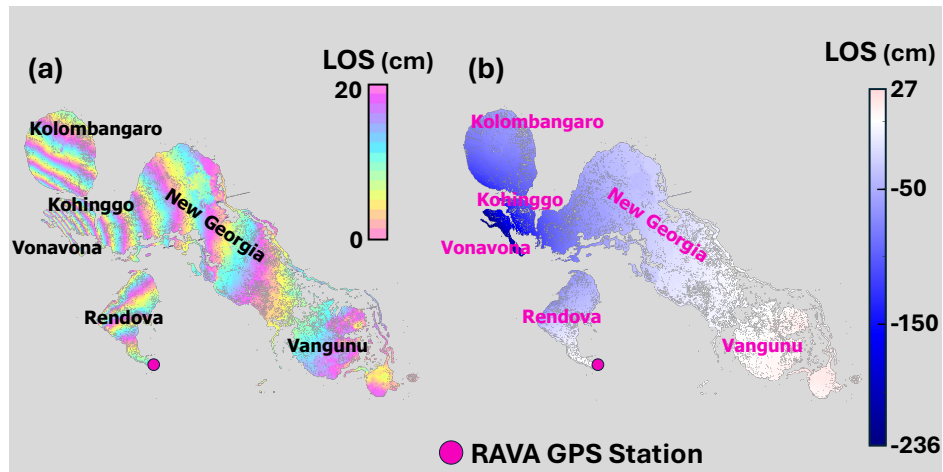


Figure 5: (a) Surface deformation associated with the 2007 Solomon Islands earthquake displayed on a circular color bar. Each fringe of corresponding colors, i.e. pink to pink, represents 20 centimeters of line-of-sight (LOS) deformation. Fringes appear denser near the epicenter. (b) The same InSAR LOS coseismic deformation map on a continuous color map. Blue (negative LOS values) indicates uplift and/or westward motion. The epicenter was identified southwest of Vonavona as shown in Figure 2.

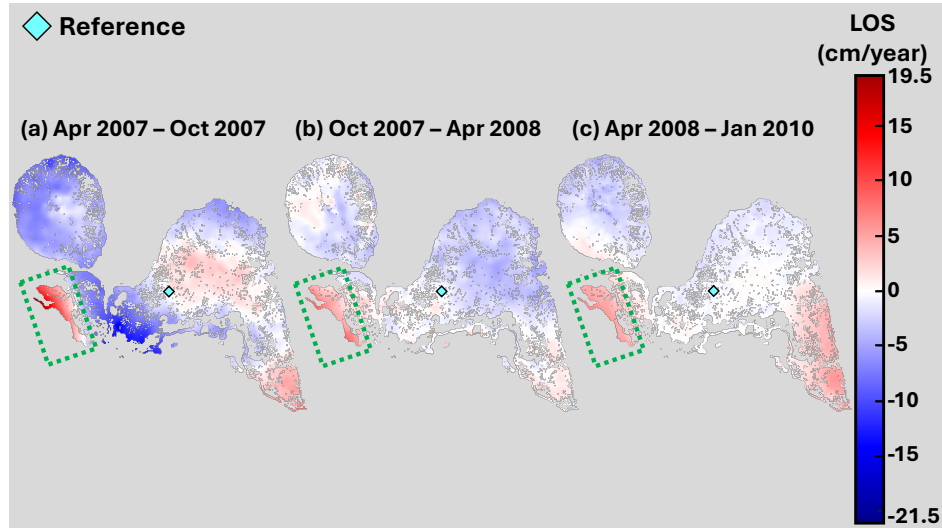


Figure 6: Average postseismic deformation rate estimates (cm/year) for the periods of (a) April 16, 2007 through October 17, 2007; (b) October 17, 2007 through April 18, 2008; and (c) April 18, 2008 through January 22, 2010. Subsidence and/or eastward motion lead to positive LOS motion (red). The LOS deformation rate is relative to the reference pixel located at 8.2°S, 157.4°E marked as a teal diamond. The results over Vonavona, highlighted within the green dashed box, reveal a consistent postseismic deformation pattern with decaying magnitude over time. Outside the green dashed box, the postseismic deformation signals are below the tropospheric noise level and thus undetectable using the ALOS PALSAR interferogram stacks.

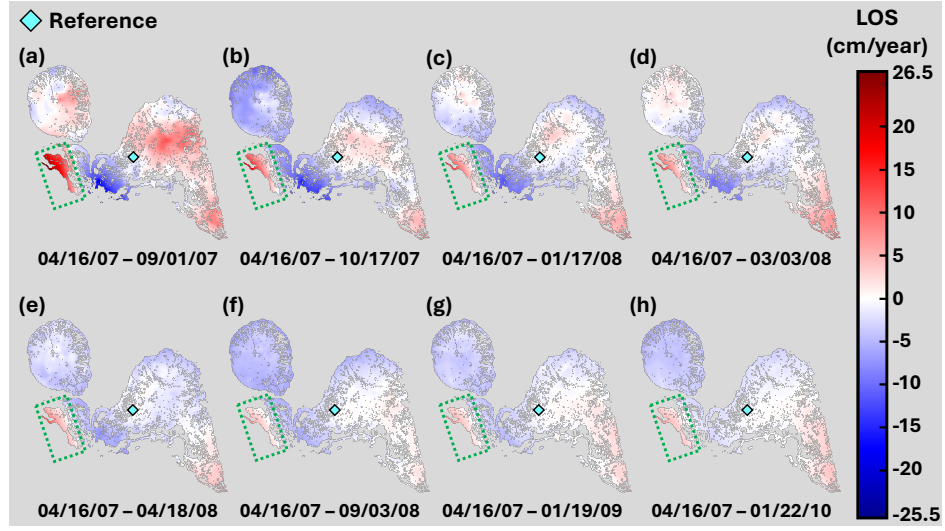


Figure 7: Average postseismic deformation rate estimates (in cm/year) for eight growing time windows starting on April 16, 2007 through: (a) September 1, 2007; (b) October 17, 2007; (c) January 17, 2008; (d) March 3, 2008; (e) April 18, 2008; (f) September 3, 2008; (g) January 19, 2009; and (h) January 22, 2010. Subsidence and/or eastward motion lead to positive LOS motion (red). The reference pixel located at 8.2°S , 157.4°E is marked by a teal diamond. Vonavona, outlined with the green dashed box, maintains a consistent postseismic deformation pattern with a magnitude that decays over time. Outside the green dashed box, the post seismic deformation signals are below the tropospheric noise level and thus undetectable using the ALOS PALSAR interferogram stacks.

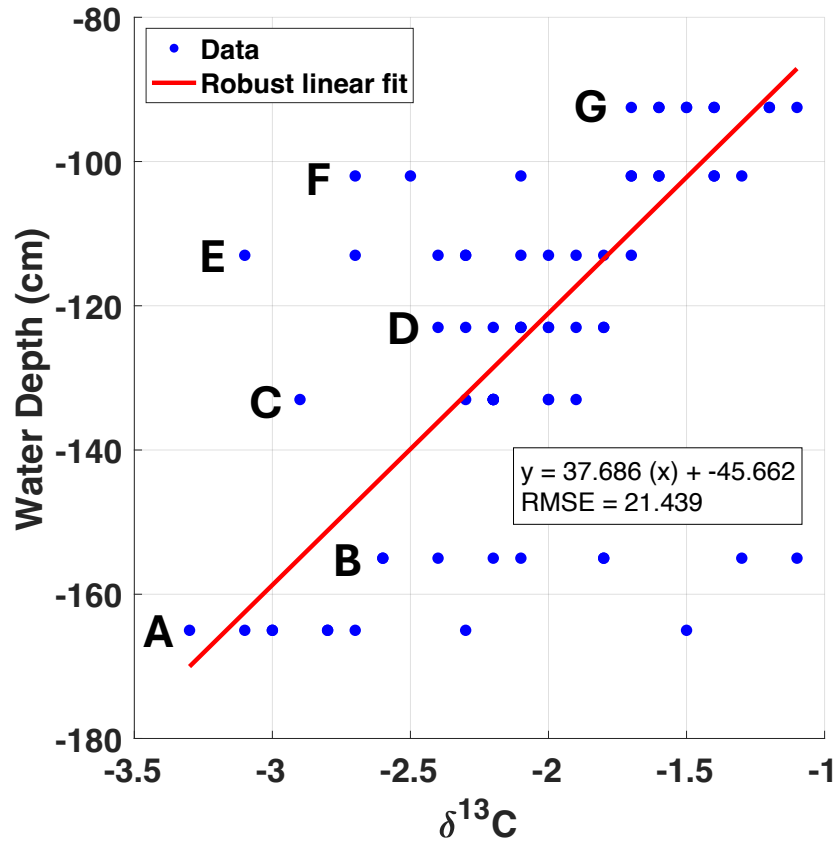


Figure 8: Transfer function between skeletal $\delta^{13}\text{C}$ and water depth. Blue dots represent $\delta^{13}\text{C}$ measurements sampled at different water depths, corresponding to the transects in Figure 4. The linear relationship (red line) is used to convert $\delta^{13}\text{C}$ measurements sampled along transect H in Figure 4 to relative sea level change time series.

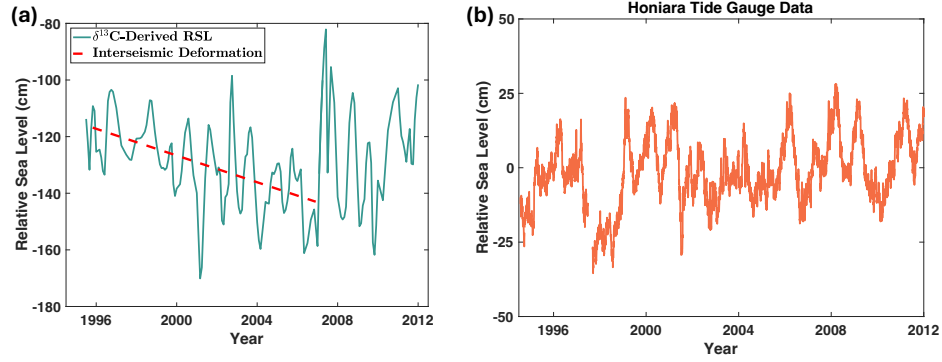


Figure 9: (a) The relative sea level change time series (in cm) as derived from coral $\delta^{13}\text{C}$ measurements. The red dashed line shows a fitted subsidence trend prior to the 2007 Solomon Islands earthquake. A large uplift signal is visible during the earthquake, which is followed by postseismic subsidence. (b) Changes in relative sea level (cm) over time captured by tide gauge data. The tide gauge data was collected at the Honiara-B station (-9.43° S, 159.96° E), which is far away from the epicenter of the 2007 Solomon Islands earthquake. Both data sets contain similar signals from climate events and seasonal weather variations, while only the coral-derived time series contains vertical tectonic motion associated with the 2007 Solomon Islands earthquake.

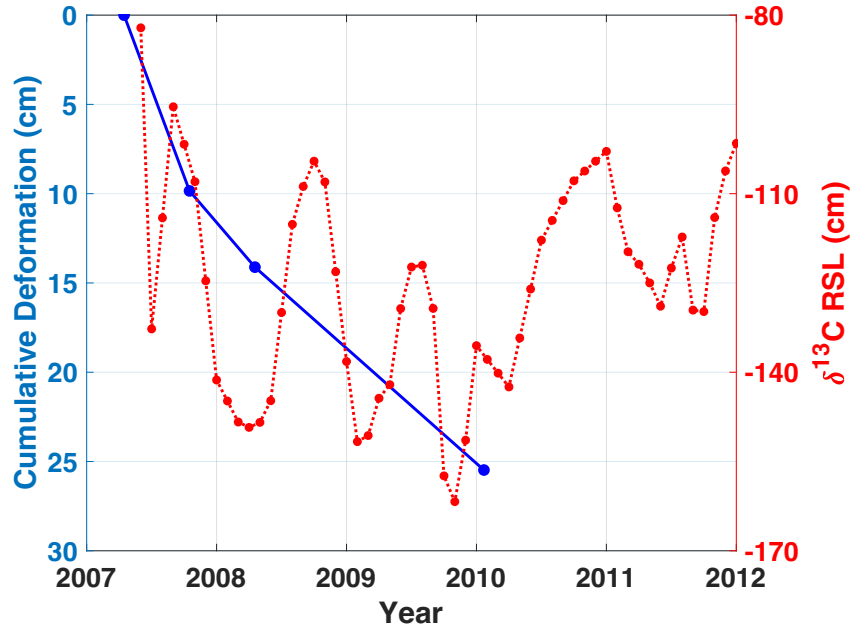


Figure 10: Total postseismic LOS deformation (cm) derived from ALOS PALSAR interferograms (blue dots). The positive sign of cumulative deformation highlights that the LOS values indicate subsidence of the land. Here the total deformation is derived from the average LOS deformation rates (cm/year) that were estimated over the following consecutive windows spanning: (1) April 16, 2007 through October 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18, 2008 through January 22, 2010 in Figure 6. Relative sea level time series derived from coral $\delta^{13}\text{C}$ measurements (red dots) show a similar decay trend.

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